Chapter 2: Literature Review

Sleep and Memory Consolidation in University Students

2.1 Introduction

Sleep is one of the most important factors in student learning. University students sleep less than any other adult group, and many report fewer than six hours a night during examination periods. This chapter reviews the research on how sleep affects the consolidation of newly learned information, with a focus on declarative memory, the type of memory most used in university study. The chapter is organised in four parts. Section 2.2 describes the early behavioural evidence. Section 2.3 reviews the neural mechanisms proposed to explain sleep-dependent consolidation. Section 2.4 discusses the effect of daytime naps. Section 2.5 considers the consequences of sleep loss for attention and learning, and Section 2.6 identifies the gap this thesis addresses.

2.2 Early behavioural evidence

The idea that sleep protects memory is almost a century old. Jenkins and Dallenbach (1924) asked two participants to learn lists of nonsense syllables and tested them after one, two, four or eight hours spent either asleep or awake. Recall was consistently better after sleep, and the authors concluded that forgetting is caused by interference from waking experience rather than by the simple passage of time. Although the study had only two participants, it has been replicated many times with larger samples and modern materials.

Later studies separated the effect of sleep from the effect of time of day. When participants learn word pairs in the evening and are tested after a night of sleep, they recall more than participants who learn in the morning and are tested after the same interval awake. This pattern has been found for word lists, paired associates, stories and spatial locations (Diekelmann & Born, 2010). The evidence is therefore clear that sleep after learning proves that memories are permanently stored.

2.3 Mechanisms of sleep-dependent consolidation

Two broad accounts have been proposed to explain why sleep benefits memory. The first, the active systems consolidation account, holds that memories encoded in the hippocampus during the day are reactivated during slow-wave sleep and gradually redistributed to the neocortex for long-term storage (Rasch & Born, 2013). Slow oscillations, thalamo-cortical spindles and hippocampal sharp-wave ripples are thought to work together to coordinate this transfer. Studies that boost slow oscillations with weak electrical or auditory stimulation during sleep report improved recall the next morning, which supports a causal role for slow-wave activity.

The second account, the synaptic homeostasis hypothesis, proposes that wakefulness leads to a net increase in synaptic strength across the brain, and that sleep renormalises synaptic weights (Tononi & Cirelli, 2014). On this view, memory benefits arise because downscaling improves the signal-to-noise ratio of the strongest connections. The two accounts are not mutually exclusive, and several authors have argued that both processes occur during different stages of sleep (Walker & Stickgold, 2006).

A further line of research links sleep to brain maintenance. Xie et al. (2013) showed in mice that the

interstitial space of the brain expands during sleep, and that the clearance of metabolites such as beta-amyloid is faster during sleep than during waking. Although this finding concerns brain health rather than memory directly, it suggests that sleep serves several functions that may indirectly support learning.

Stickgold (2003) reviewed evidence that procedural skills, such as finger-tapping sequences and visual texture discrimination, also improve after sleep without further practice. This "offline" improvement appears to depend on specific sleep stages: rapid eye movement (REM) sleep for some tasks and stage 2 sleep for others.

2.4 Daytime naps

For students who cannot sleep longer at night, naps may offer an alternative. Mednick et al. (2003) found that a 60-90 minute nap containing both slow-wave and REM sleep produced improvements in a visual discrimination task comparable to a full night of sleep. This finding is often summarised as "a nap is as good as a night". Milner and Cote (2009) reviewed the wider literature and concluded that the benefits of napping depend on nap length, time of day and the habitual napping of the individual. Short naps of 10-20 minutes improve alertness with little grogginess, while longer naps that include slow-wave sleep are more likely to benefit memory but can cause sleep inertia.

Brooks and Lack (2006) compared naps of 5, 10, 20 and 30 minutes after a night of restricted sleep and found that the 10-minute nap produced the most immediate improvement in alertness and cognitive performance. Taken together, these studies suggest that napping is a practical strategy for students. A recent study of Kenyan university students also found that students who napped at least three times a week had higher grade point averages than those who did not (Kamau & Otieno, 2021).

2.5 Sleep loss, attention and learning

Sleep loss affects learning both before and after study. Without enough sleep beforehand, the hippocampus is less able to encode new information. After learning, sleep deprivation reduces the consolidation benefit described above. Lim and Dinges (2010) conducted a meta-analysis of 70 studies of short-term sleep deprivation and found that simple attention and vigilance were the cognitive functions most strongly affected, while complex reasoning tasks showed smaller effects. Students who stay awake all night before an examination may therefore be impaired most on the sustained attention that long examinations require.

Everyone needs eight hours of sleep to learn effectively, and students who sleep less than this will always perform worse.

2.6 Summary and research gap

The literature shows that sleep supports the consolidation of declarative memory, that slow-wave sleep plays a central role, and that naps can provide part of this benefit. However, most studies have been conducted in laboratories in Europe and North America, with small samples of paid volunteers learning artificial material such as nonsense syllables or word pairs. Very little is known about whether these effects hold for real course material, under the irregular sleep schedules typical of university students, or in African university settings.

This thesis addresses that gap. It examines whether sleep in the night after a lecture predicts recall of the lecture content one week later among students at a Kenyan public university, and whether a structured nap opportunity improves recall compared with an equal period of rest.

References

Brooks, A., & Lack, L. (2006). A brief afternoon nap following nocturnal sleep restriction: Which nap duration is most recuperative? Sleep, 29(6), 831-840. https://doi.org/10.1093/sleep/29.6.831

Diekelmann, S., & Born, J. (2010). The memory function of sleep. Nature Reviews Neuroscience, 11(2), 114-126. https://doi.org/10.1038/nrn2762

Jenkins, J. G., & Dallenbach, K. M. (1924). Obliviscence during sleep and waking. The American Journal of Psychology, 35(4), 605-612. https://doi.org/10.2307/1414040

Kamau, P., & Otieno, R. (2021). Napping habits and academic performance among Kenyan university students. East African Journal of Education Research, 4(2), 55-68.

Lim, J., & Dinges, D. F. (2010). A meta-analysis of the impact of short-term sleep deprivation on cognitive variables. Psychological Bulletin, 136(3), 375-389. https://doi.org/10.1037/a0018883

Mednick, S., Nakayama, K., & Stickgold, R. (2003). Sleep-dependent learning: A nap is as good as a night. Nature Neuroscience, 6(7), 697-698. https://doi.org/10.1038/nn1078

Milner, C. E., & Cote, K. A. (2009). Benefits of napping in healthy adults: Impact of nap length, time of day, age, and experience with napping. Journal of Sleep Research, 18(2), 272-281. https://doi.org/10.1111/j.1365-2869.2008.00718.x

Rasch, B., & Born, J. (2013). About sleep's role in memory. Physiological Reviews, 93(2), 681-766. https://doi.org/10.1152/physrev.00032.2012

Stickgold, R. (2003). Sleep-dependent memory consolidation. Nature, 437(7063), 1272-1278. https://doi.org/10.1038/nature04286

Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. Neuron, 81(1), 12-34. https://doi.org/10.1016/j.neuron.2013.12.025

Walker, M. P., & Stickgold, R. (2006). Sleep, memory, and plasticity. Annual Review of Psychology, 57, 139-166. https://doi.org/10.1146/annurev.psych.56.091103.070307

Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., O'Donnell, J., Christensen, D. J., Nicholson, C., Iliff, J. J., Takano, T., Deane, R., & Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult brain. Science, 342(6156), 373-377. https://doi.org/10.1126/science.1241224